

1. Concept of Function Composition:

A. $y = k(h(g(x))) \equiv y = k \circ h \circ g(x).$

B. presence of neural network is composite function.

2. Concept of Chain Rule:

A. $y = k \circ h \circ g(x) \longrightarrow y' = \frac{\partial k}{\partial x} = \frac{\partial k}{\partial h} \cdot \frac{\partial h}{\partial g} \cdot \frac{\partial g}{\partial x}$

B. backpropagation need to use chain rule to compute composite function in neural network.